Phase 5:Project Development Phase

SmartLender: AI-Powered Credit Card Approval System

Role	Name
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Implementation Overview

Development followed the design laid out in Phase 3. Data cleaning, feature engineering, and model training were carried out in a Jupyter notebook. The trained model, scaler, and encoders were serialized and loaded into a Flask application that exposes a form-based prediction endpoi

Key Modules

- Training/ – data preparation, EDA, model training and tuning notebook
- Flask/app.py – request handling and prediction endpoint
- Flask/templates/index.html – applicant input form
- Flask/model/ – serialized model, scaler, and encoders

Model Results

Logistic Regression, Decision Tree, Random Forest, and XGBoost were trained and compared. XGBoost was tuned further with RandomizedSearchCV and selected as the production model based on its cross-validated accuracy.

Model	Accuracy
Logistic Regression	0.762
Decision Tree	0.741
Random Forest	0.799
XGBoost (tuned, 5-fold CV)	**0.811**

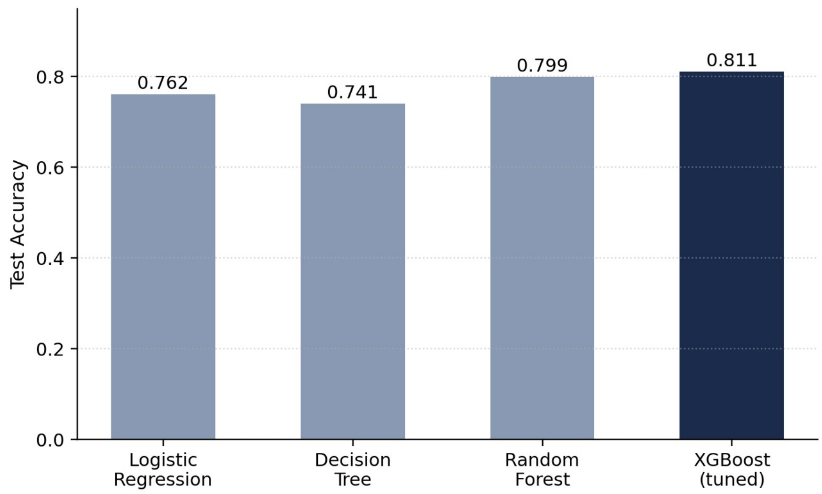

Figure 4: Model accuracy comparison